Experiment 8 - Conduct Z test for the Data given using Python

Aim:

To conduct a one-sample Z-test in Python to test whether the average weight of a species of birds differs from the claimed population mean of 150 grams.

Program:

```
import numpy as np
import scipy.stats as stats

sample_data = np.array([152, 148, 151, 149, 147, 153, 150, 148, 152, 149,151, 150, 149, 152,
151, 148, 150, 152, 149, 150,148, 153, 151, 150, 149, 152, 148, 151, 150, 153])
population_mean = 150
sample_mean = np.mean(sample_data)
sample_std = np.std(sample_data, ddof=1)

n = len(sample_data)
z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))
p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic)))

print(f"Sample Mean: {sample_mean:.2f}")
print(f"Z-Statistic: {z_statistic:.4f}")
print(f"P-Value: {p_value:.4f}")
alpha = 0.05

if p_value < alpha:
    print("Reject the null hypothesis: The average weight is significantly different from 150
grams.")
else:
    print("Fail to reject the null hypothesis: There is no significant difference in average
weight from 150 grams.")
```

Output:

```
Sample Mean: 150.20
Z-Statistic: 0.6406
P-Value: 0.5218
Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.
```

Result:

Thus, a one-sample Z-test in Python to test whether the average weight of a species of birds differs from the claimed population mean of 150 grams is conducted.